English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

Computer Science, 30 June 2024 Examination *The grade is 2 + the number of points. Time allowed: 4 hours. Good luck!*

Problem 1. Asen and five friends play football daily, choosing uniformly at random two teams of three. Find the expectation and variance of the number of days until:

1. (0.25 points) Asen first plays with his two best friends;
2. (0.25 points) a previously used team division first recurs;
3. (0.5 points) all possible team divisions have occurred.

Problem 2. Boris and Valya arrange to meet outside FMI. Their travel times are independent exponential variables with mean 10 minutes.

1. (0.25 points) What is the probability, to accuracy 0.01 if desired, that the first arrives in under 10 minutes and the other one minute later?
2. (0.25 points) Find x such that the probability that both take at least x minutes is 50%.
3. (0.25 points) After 100 such arrangements, approximately what is the probability, to accuracy 0.01, that Boris arrived first at least 55 times?
4. (0.5 points) Find the expectation, variance and density of the waiting time of the first arrival until the second arrives.

Problem 3.

1. (0.5 points) Let Ω have 10 elements. In how many ways can three subsets A, B, C satisfy
$$\mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C) = 1 + \mathbb{P}(A \cap B) + \mathbb{P}(A \cap C) + \mathbb{P}(B \cap C)?$$
2. (0.5 points) Choose uniformly a nondecreasing function from $\{1, \dots, 10\}$ to $\{1, \dots, 20\}$. What is the probability that it is strictly increasing?

Problem 4. Let X, Y be independent Exp(1) variables and $Z = \sqrt{X/Y}$.

1. (0.5 points) Find F_Z and use it to calculate $\mathbb{E}Z$.
2. (0.5 points) Express $\mathbb{E}Z$ using the joint density of X, Y. Deduce $\Gamma(1/2) = \sqrt{\pi}$.[1]

Bonus (0.5 points): Find $\lim_{n\to\infty} e^{-n}\left(\frac{n}{1!} + \frac{n^2}{2!} + \cdots + \frac{n^n}{n!}\right)$.

[1] Euler's gamma function is $\Gamma(z) = \int_0^\infty t^{z-1}e^{-t}\,\mathrm{d}t$ and satisfies $\Gamma(z+1) = z\Gamma(z)$.